Explore Numerical Expressions with Exponents

Write a power as repeated multiplication and find its value.

Name _____ **Date** _____

Tonight's spark

A school phone tree starts when you text 3 friends. Then each new person texts 3 more friends. In round 4, how many people get the text?

Make a guess first: more than 50 people, or fewer?

Round	People
1	3
2	9
3	27
4	?

Our class model: write it out as copies

Try it

Work it out on paper. Write one answer on each line.

1 **How many factors of 8 are multiplied in 8^5?**

A. 8 **B.** 5 **C.** 40 **D.** 13

My answer _____

2 **What is the value of 2 × 2 × 2 × 2 × 2?**

A. 32 **B.** 10 **C.** 25 **D.** 16

My answer _____

3 **What is the value of 5^3?**

A. 15 **B.** 53 **C.** 125 **D.** 25

My answer _____

4 **Which power has the greatest value?**

A. 3^2 **B.** 2^3 **C.** 1^5 **D.** 2^4

My answer _____

5 **What is the value of 7^2?**

A. 14 **B.** 49 **C.** 72 **D.** 9

My answer _____

6 **A scientist writes 10^4 for the number of cells. How many cells is that?**

A. 10,000 **B.** 40 **C.** 1,000 **D.** 100,000

My answer _____

Talk about it

How many copies of the base did you multiply, and how do you know from the exponent?

Home mission

Use coins or buttons. Put out 3 for round 1. For round 2, add 3 new ones for each one. Count. Then say the powers aloud: 3 to the 1st, 2nd, 3rd. Predict round 4 before you build it.

If we get stuck, ask

- Can you write this out as repeated multiplication first?
- How many copies of the base are there?
- Are we multiplying BY the small number, or multiplying copies together?

Do it on a screen

Scan or type the link. Same lesson, with hints.

eduwonderlab.com/lessons/6-3-part2/homework.html

How did it feel? (optional)

Tricky Getting there Got it